

GoMart Search Analysis

Sep 2020

Background

Problem Statement: Evaluate Relevance of GoMart Search

OKR Increase Visit to Booking to 20%

Context: There is a ~**52% drop** in booking funnel from **Search Result Shown to Add Item**

Data

Event: SKU Item Added

Source: Search Screen

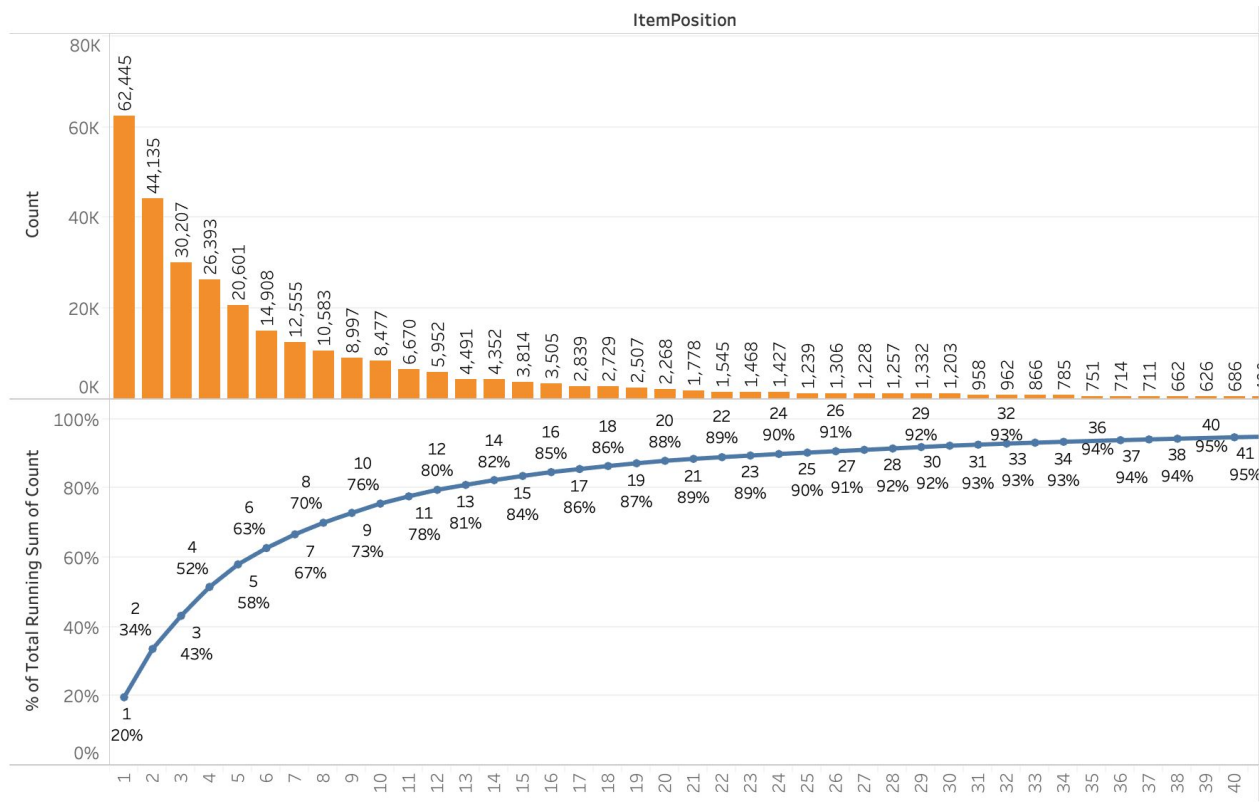
1 Month from '2020-08-23' and '2020-09-22'

Data is available only for iOS

For Android, wrong source was getting added when sku item was added from search screen

This will be corrected in the upcoming release 4.6. [Details](#)

Distribution of Item position for SKUs added from Search Screen



- **43%** of the SKUs are added from the **Top 3** positions in search
- **63%** of the SKUs are added from the **Top 6** positions in search
- **80%** of the SKUs are added from the **Top 12** positions in search
- **90%** of the SKUs are added from the **Top 24** positions in search
- Rest additions forms a very long tail

[Detailed List](#)

Next Approach

We Identify the **Top 50 search terms**, where items are added from **position > 4**

For these search terms, we get the distribution of position from which the skus were eventually added

Position sets are created **1-6, 7-12, 13-18, 19-24, 25+**

We classify Search Terms into two broad categories

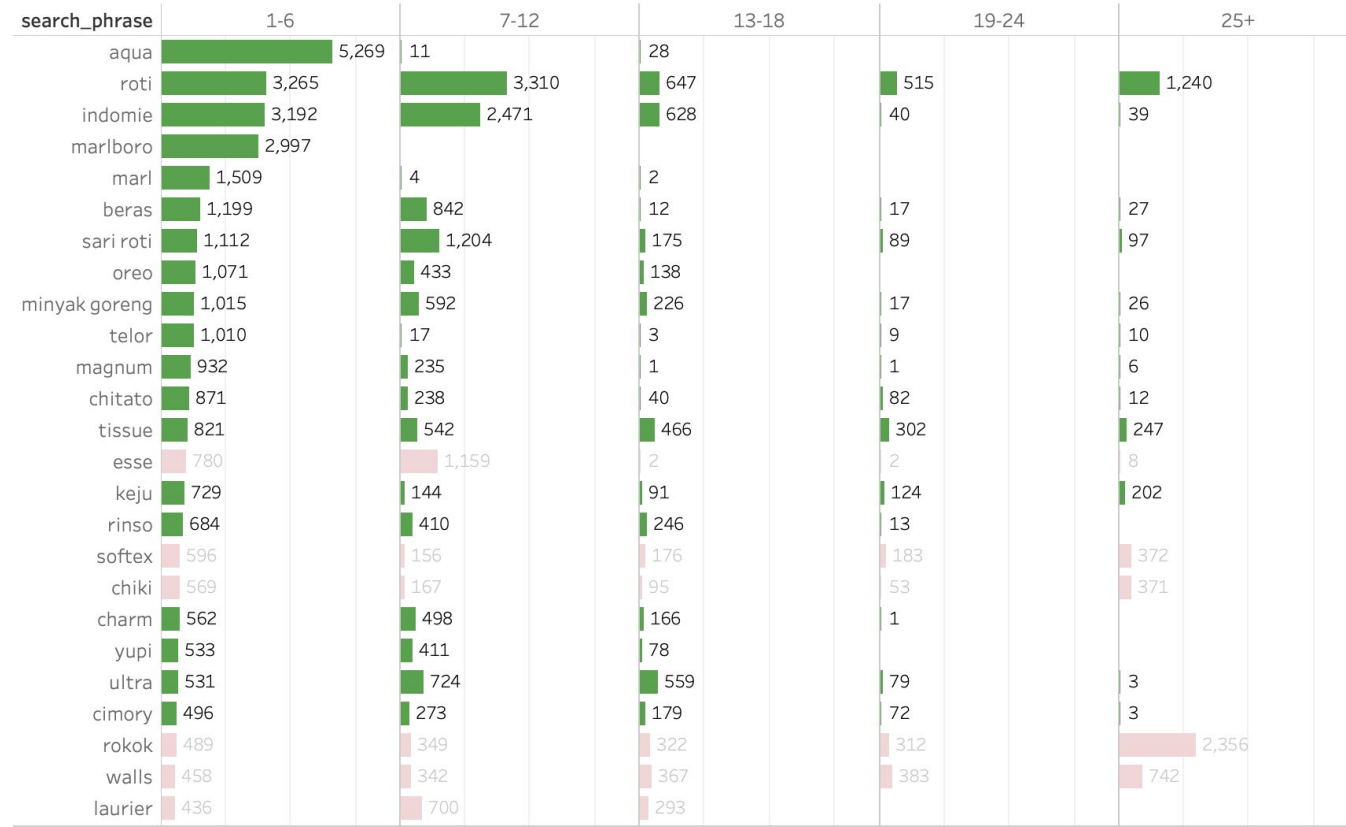
- High Relevance
- Low Relevance

High relevance search terms have higher selections in top positions

1. High Relevance

Search terms for which high proportion of skus are added from initial position set in comparison to next position set

These are further classified into two categories based on assortment



Searches for exact product names like *aqua*, *marlboro*, *oreo* results in high relevance

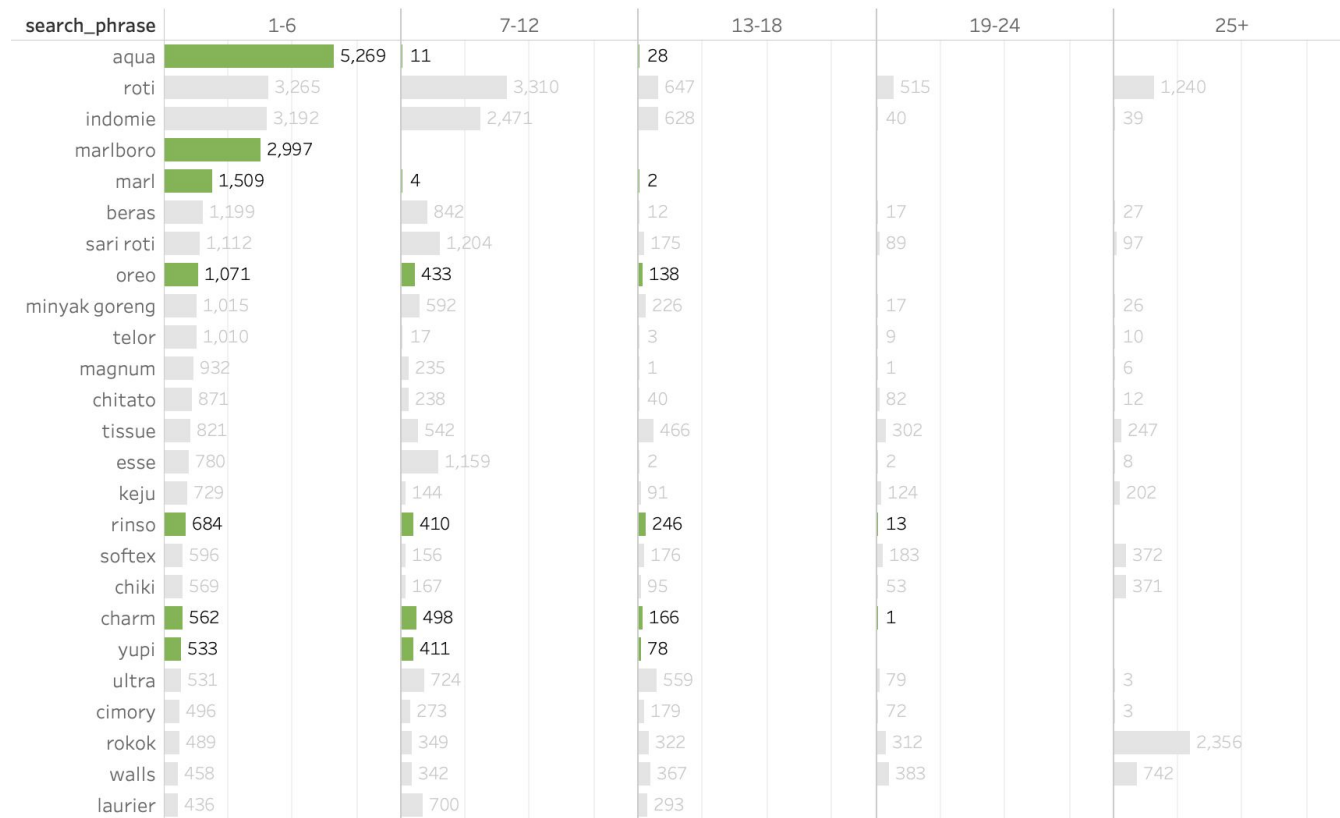
1.a High Relevance Low Assortment

These are the terms from which all item additions are from top 12-18 search results

Examples:

Aqua, marlboro, marl, oreo, risno, charm, yupi

Search terms are branded items or exact product names



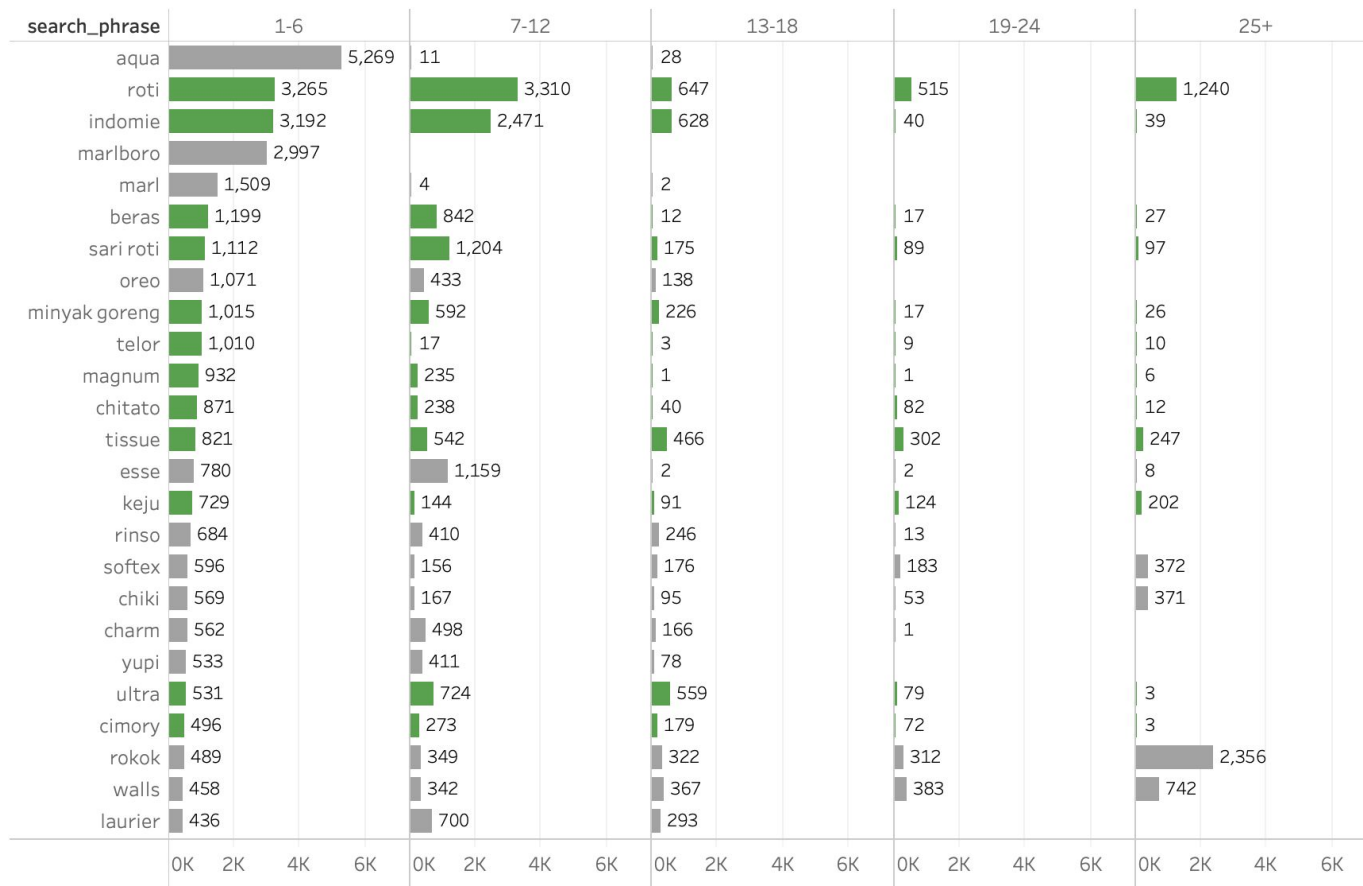
Roti, indomie, tissue have high relevance with long tail of selection due to high assortment

1.b High Relevance High Assortment

These are the terms from which high proportion of items are added from top 12-18 search results and have a long tail due to wide assortment

Examples:

Roti, indomie, beras, tissue, ultra, cimory



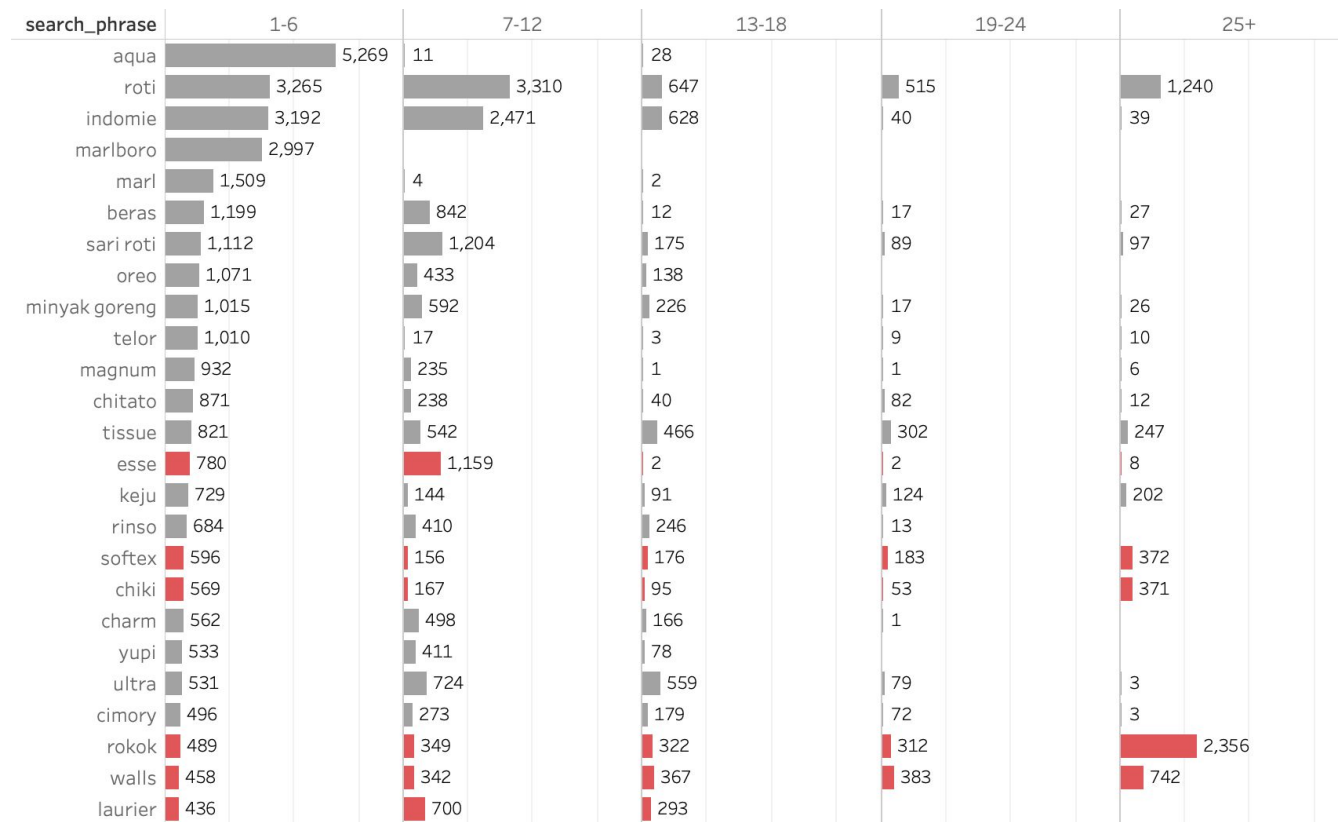
Large brands names like *esse*, *softex*, *walls* having high product portfolio suffer from low relevance

2. Low Relevance

These are the search terms from which high proportion of items are added from a later position set in comparison to previous position set

Examples:

Esse, softex, chiki, rokok, walls



Low relevance is also seen for generic searches like *susu*, *snack*, *ice cream*

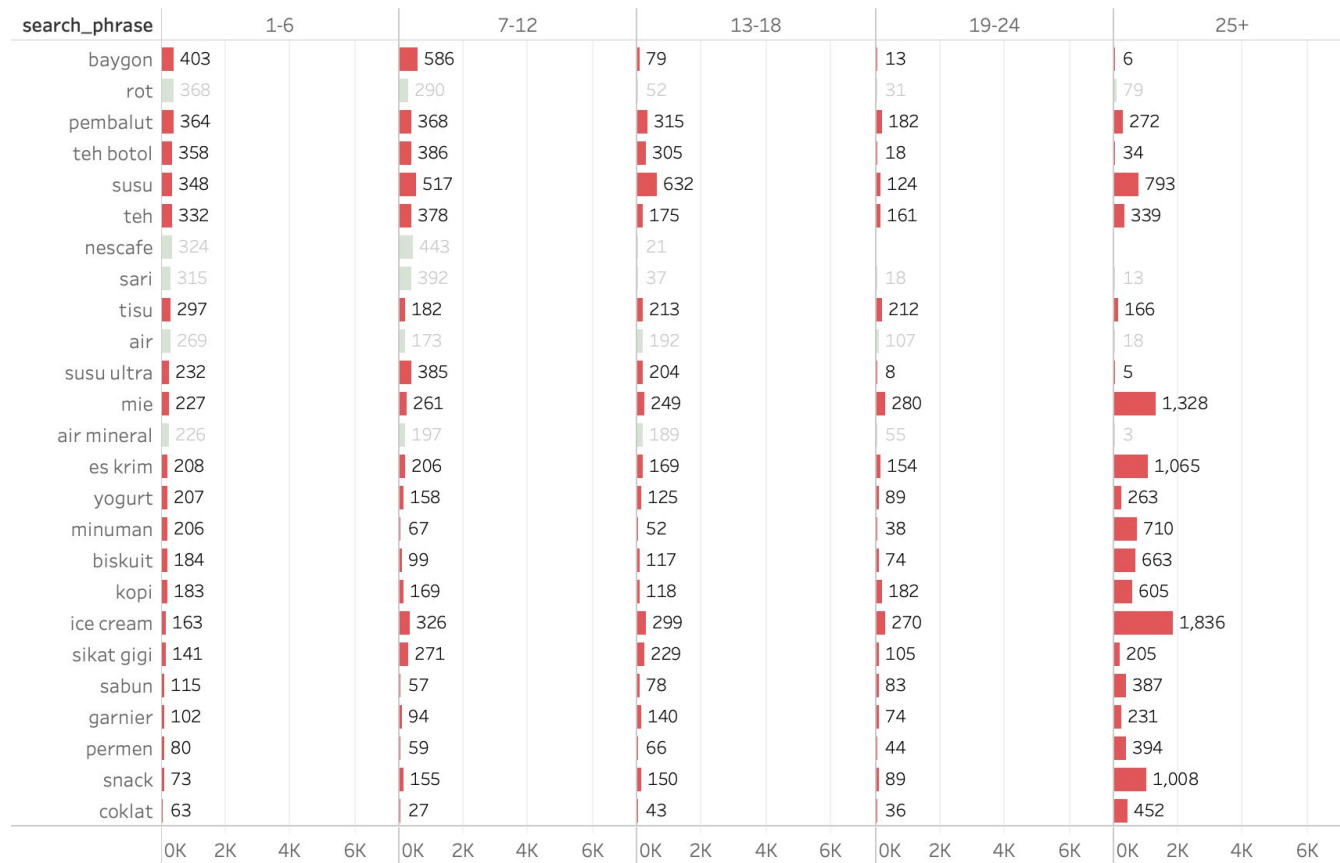
2. Low Relevance (contd)

Next set of 25 search terms

Examples:

Baygon, susu, tisu, mie, es krim, yogurt, ice cream, snack, coklat

These search terms are category /sub category searches and not exact product name with brand



SKU Item Added Based on Category

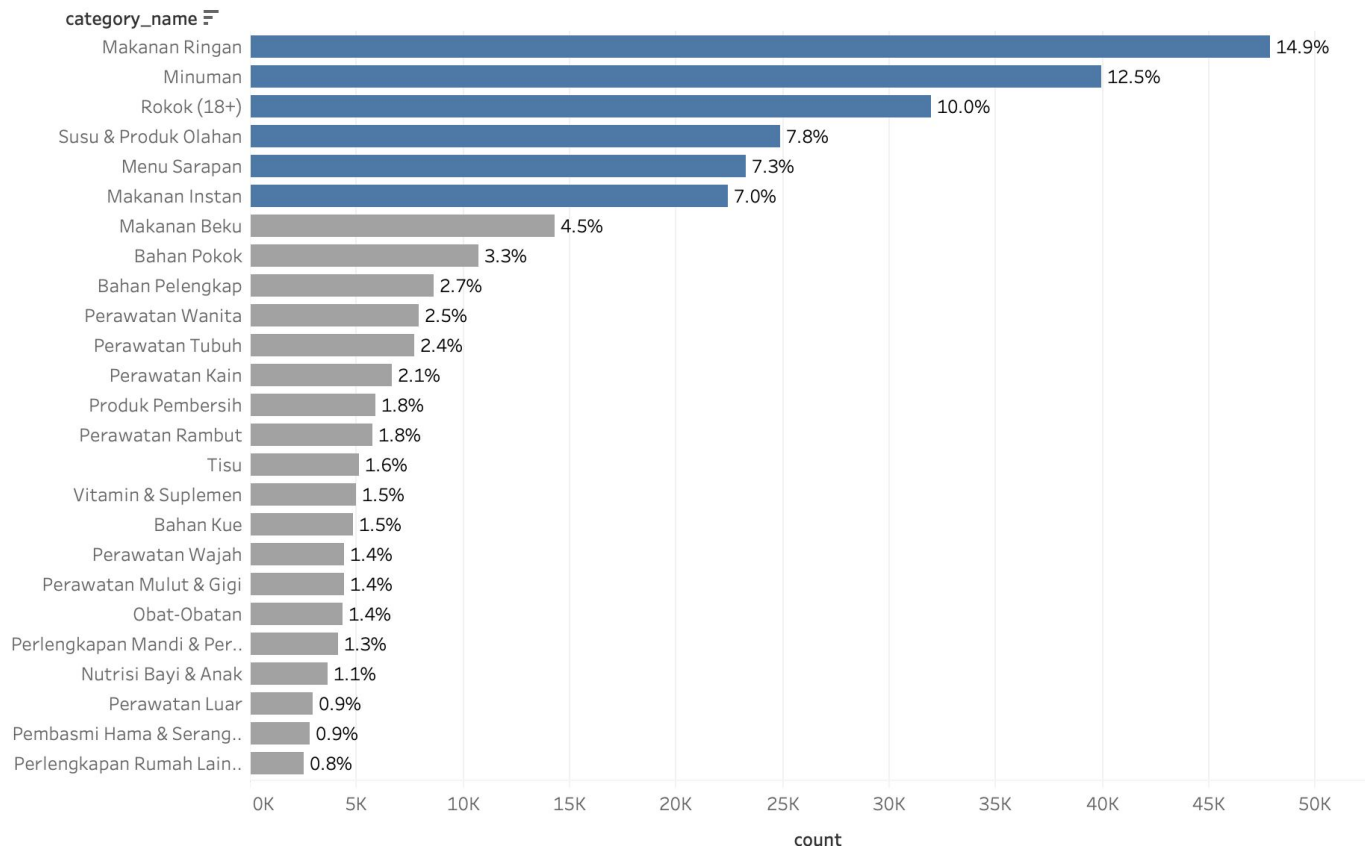
CT Analysis - From iOS only

SKU Item Added

In the last 30 days

- In the last 30 day(s)
- **ServiceType** contains "Go-Buy" ✗
- AND **Source** contains "Search Screen" ✗

Popular Search categories are Makanan Ringan, Minuman, Rokok



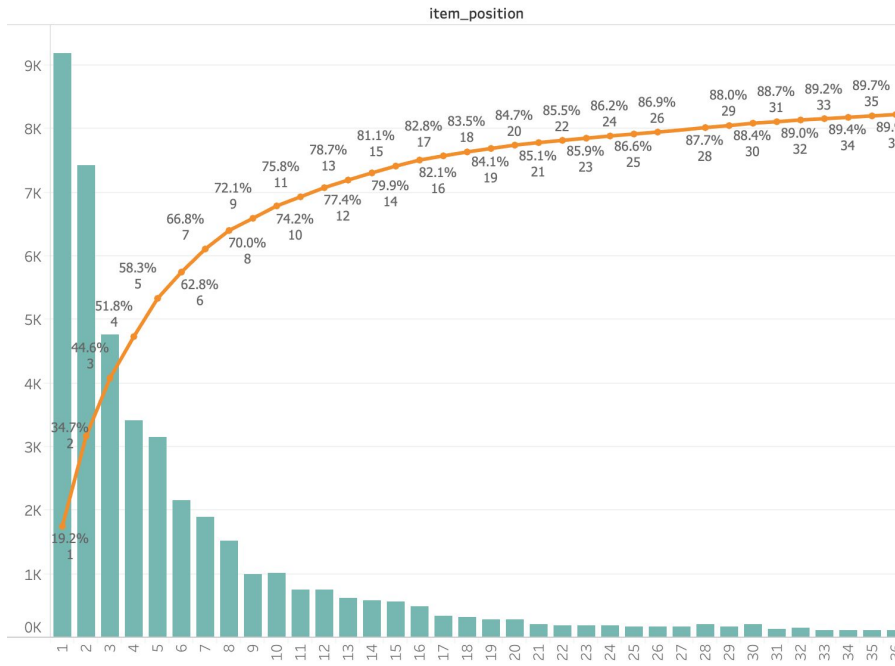
60% of items are added from the Top 6 Categories

Rest 40% of items are added from other 57 categories

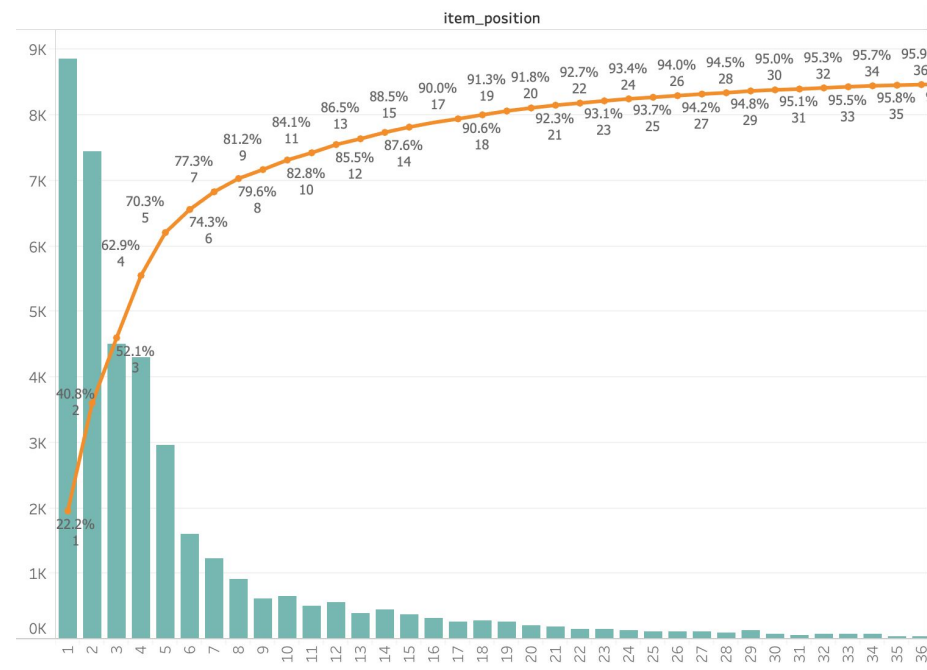
[Detailed List](#)

Distribution of Categories by Position

Makanan Ringan



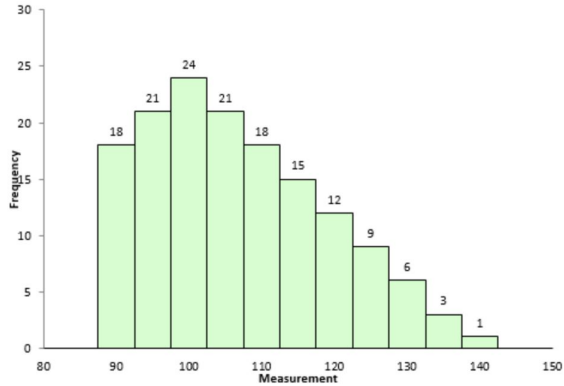
Minuman



The distribution is better analysed by looking at skewness for every category
Ideally we would want positive skewed distribution

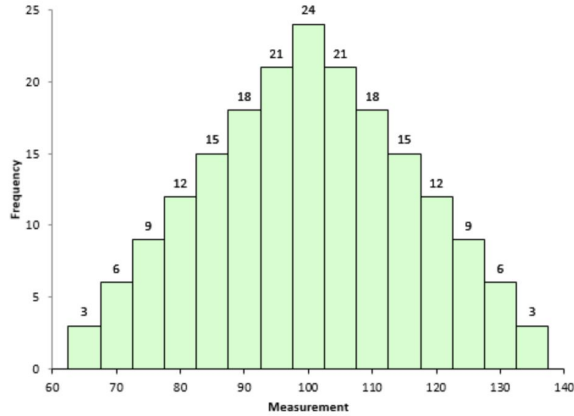
Skewness

Figure 2: Dataset with Positive Skewness



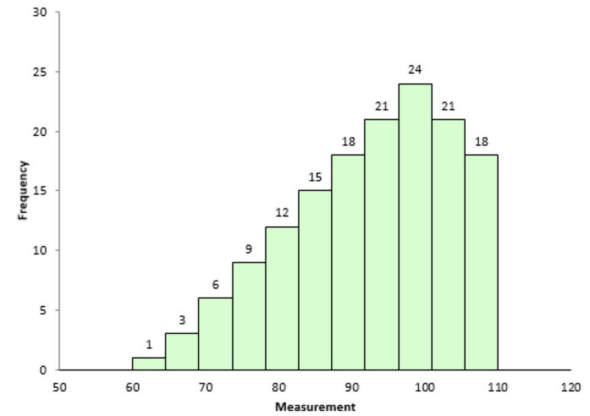
Skewness : 0.514
Excess Kurtosis : -0.527

Figure 1: Symmetrical Dataset with Skewness = 0



Skewness : 0.0
Excess Kurtosis : 0.0

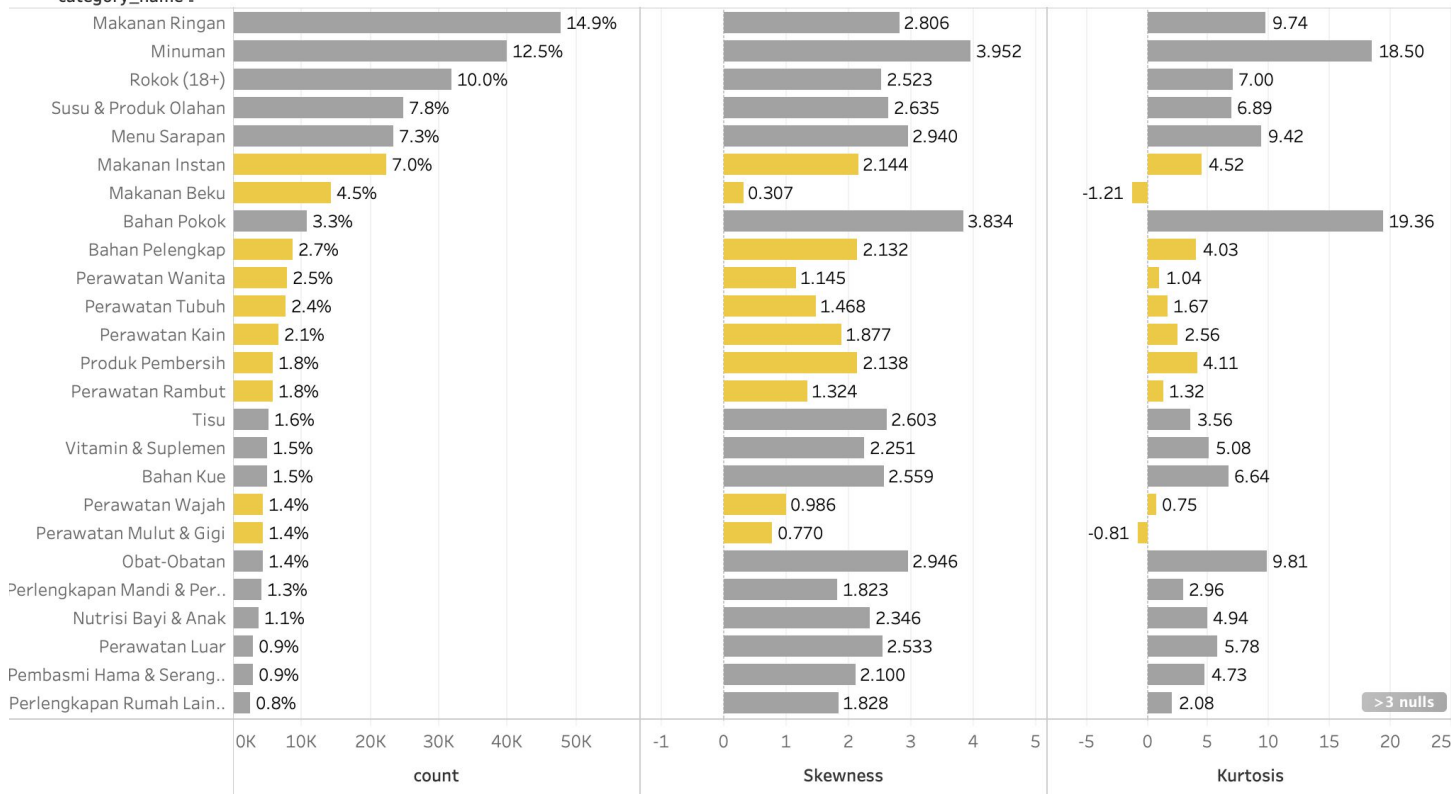
Figure 3: Dataset with Negative Skewness



Skewness : -0.514
Excess Kurtosis : -0.527

Search relevance from *Makanan Beku*, *Perawatan Wanita*, *Perawatan Rambut*, *Perawatan Tubuh* is less

category_name



Based on the **skewness** and **count** we identify categories where the distribution is not positively skewed

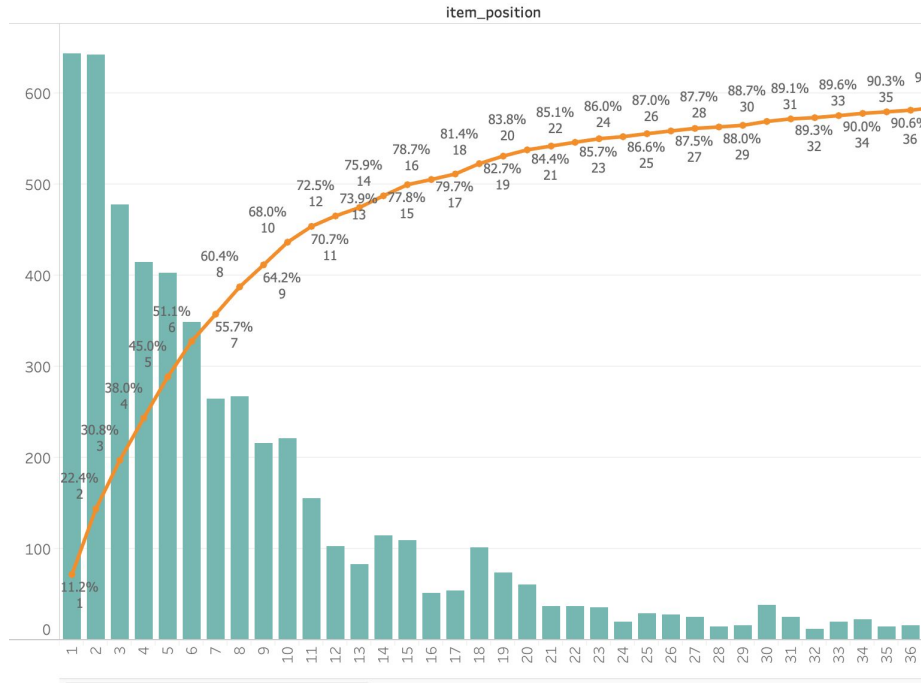
Users have to scroll more to add items which are from these categories, in comparison to other categories

Examples:

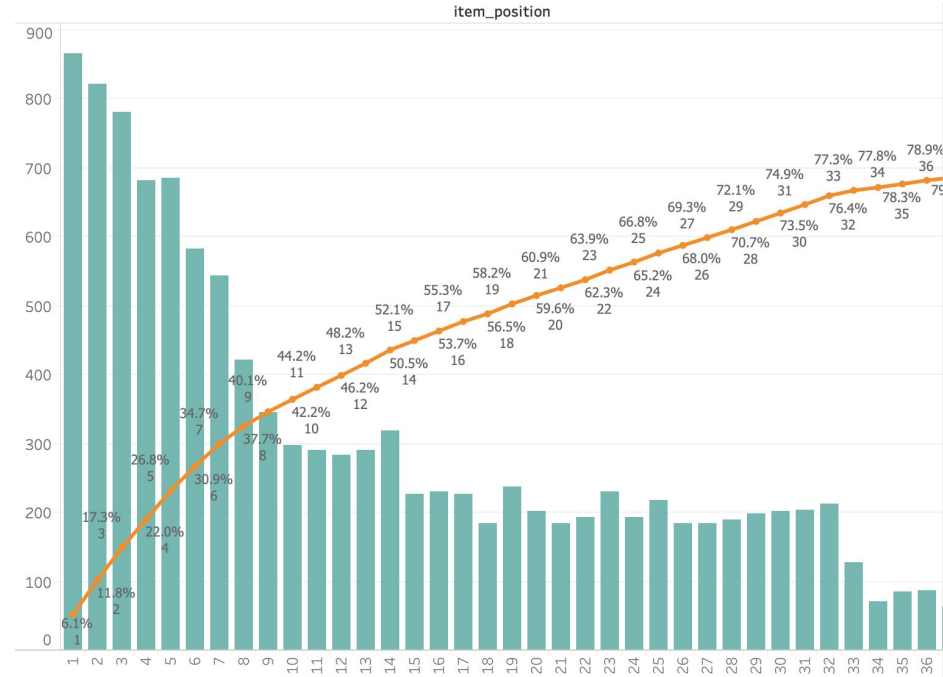
Makanan Beku, Perawatan Wanita, Perawatan Rambut, Perawatan Tubuh

Distribution of Less Positive Skewed Categories

Perawatan Rambut



Makanan Beku

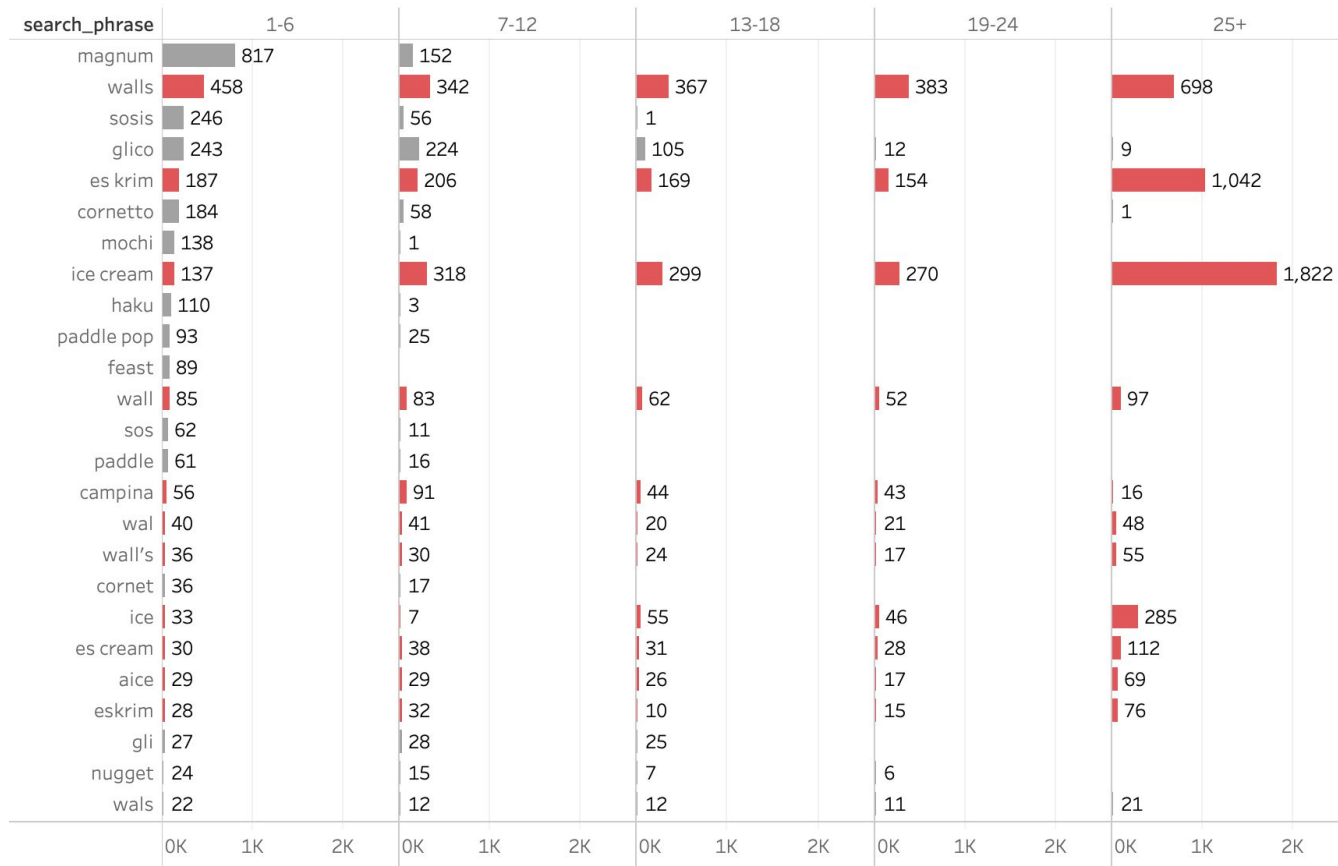


Low Relevance - Makanan Beku

Search terms with low relevance
in makanan beku category

Examples:

Wall, es krim, ice cream, ice, aice,
eskrim



Low Relevance - Perawatan Rambut

Search terms with low relevance
in perawatan rambut category

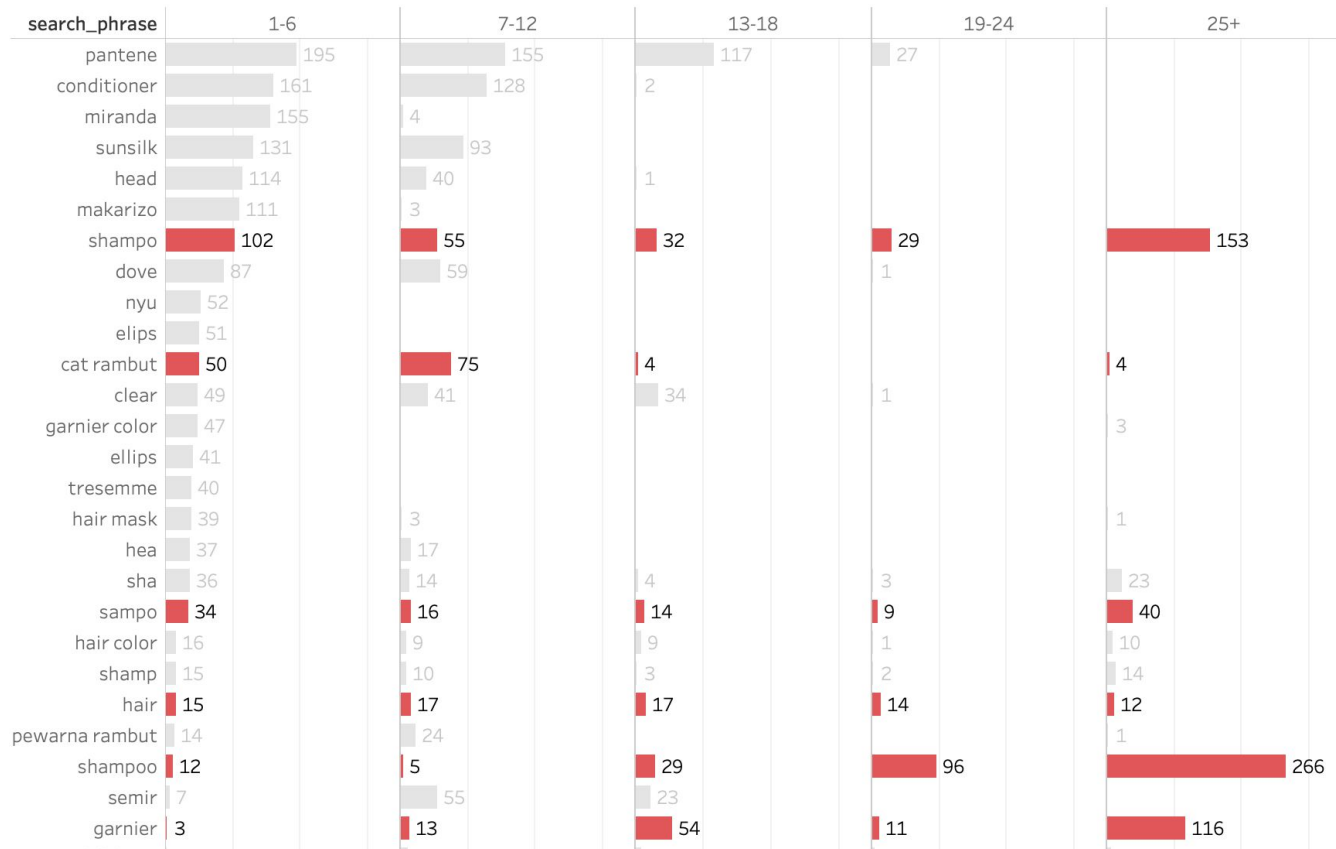
Examples:

Shampoo, cat rambut, sampo,
hair, shampoo, garnier

For branded search, relevance is
high. Eg. pantene, sunsilk

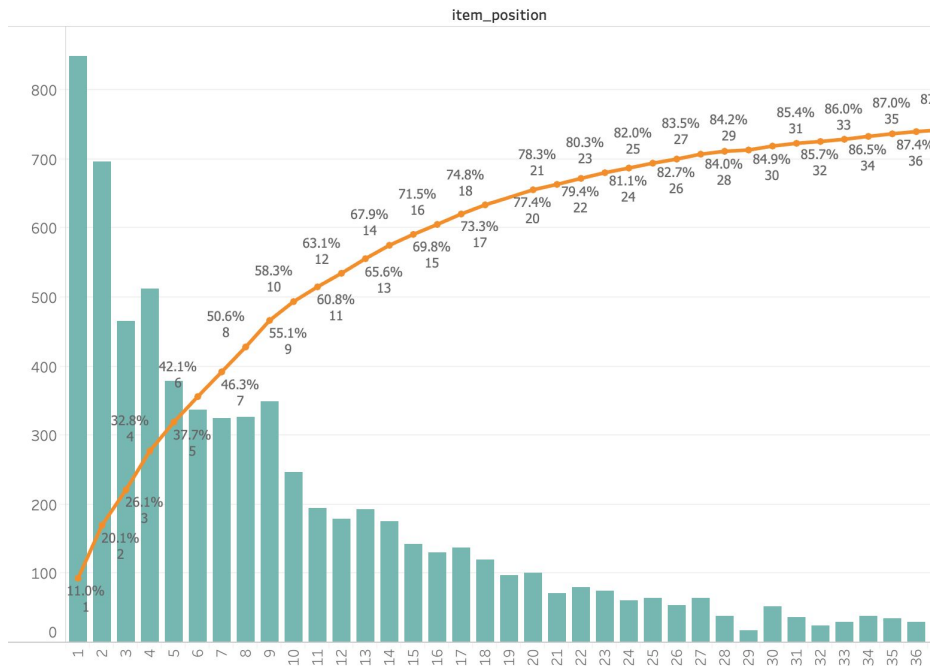
For generic searches the
relevance decreases

Eg. shampoo, garnier, sampo

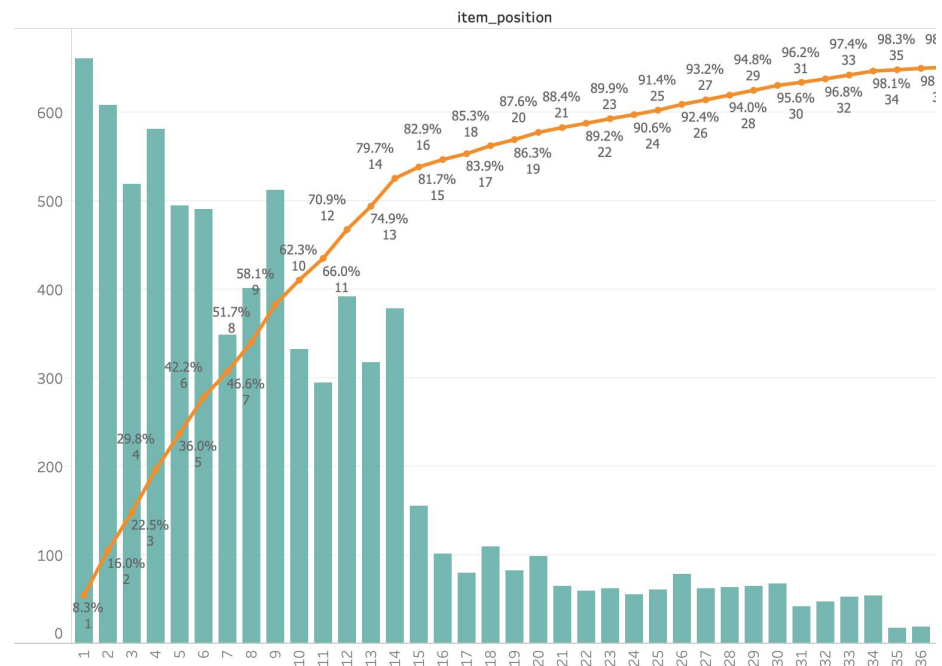


Distribution of Less Positive Skewed Categories

Perawatan Tubuh



Perawatan Wanita

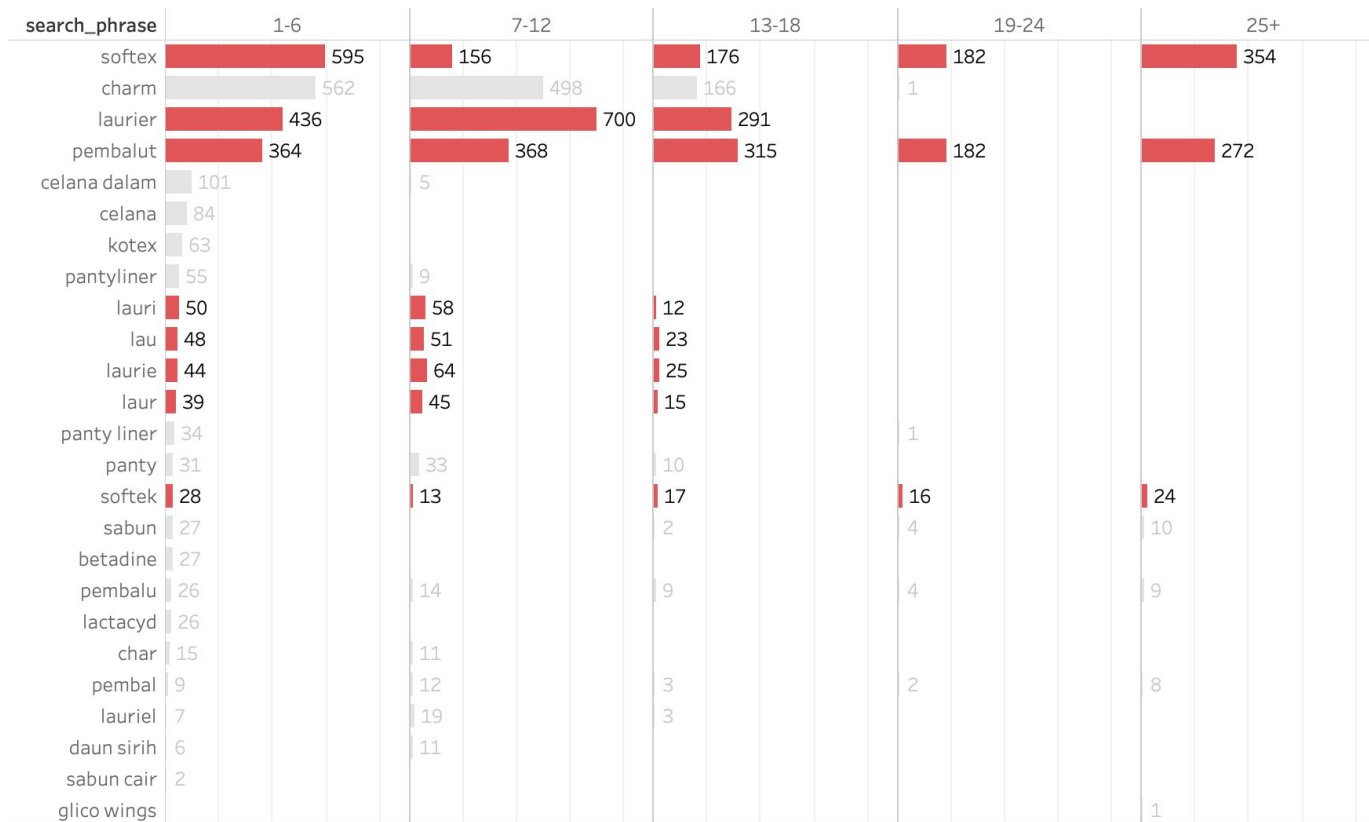


Low Relevance - Perawatan Wanita

Search terms with low relevance
in perawatan wanita category

Examples:

Softex, pembalut, laurier

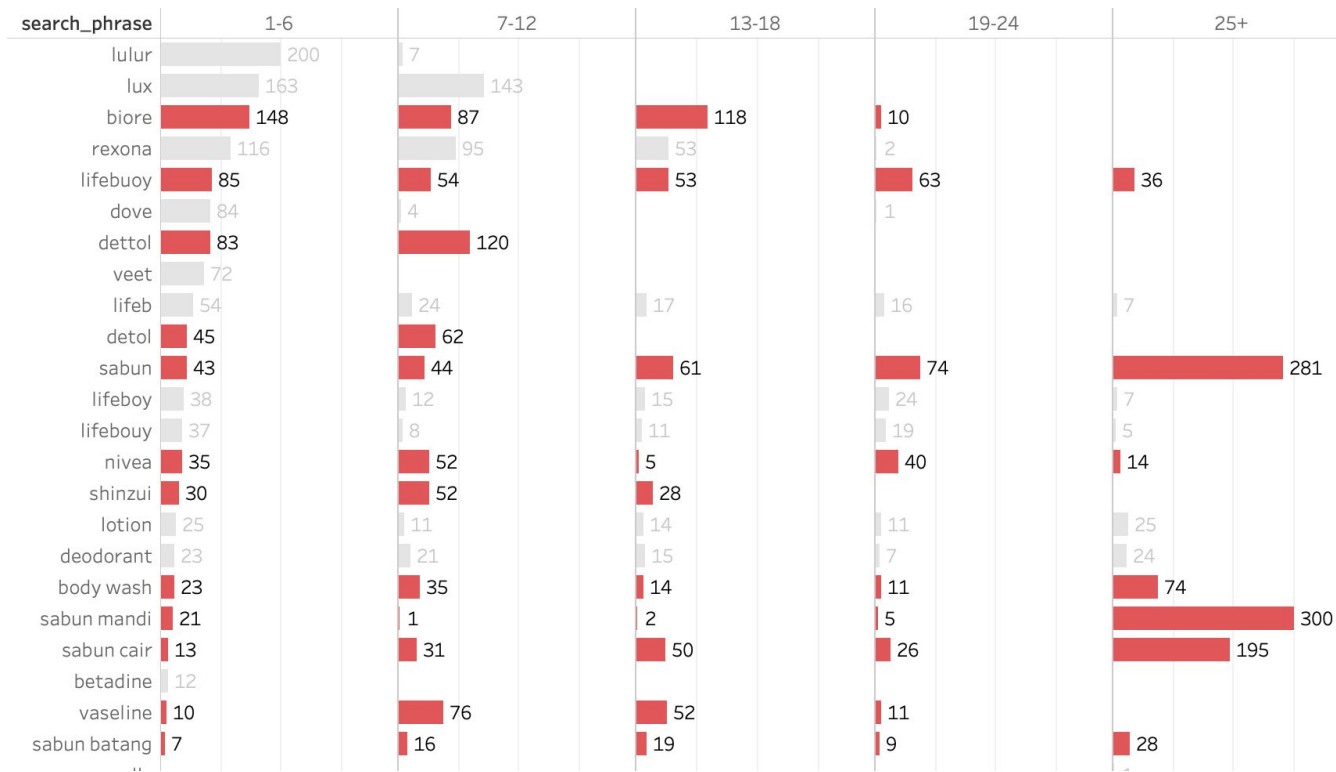


Low Relevance - Perawatan Tubuh

Search terms with low relevance
in perawatan tubuh category

Examples:

Sabun, sabun mandi, sabun cair,
body wash



SKU Item Added Based on Items

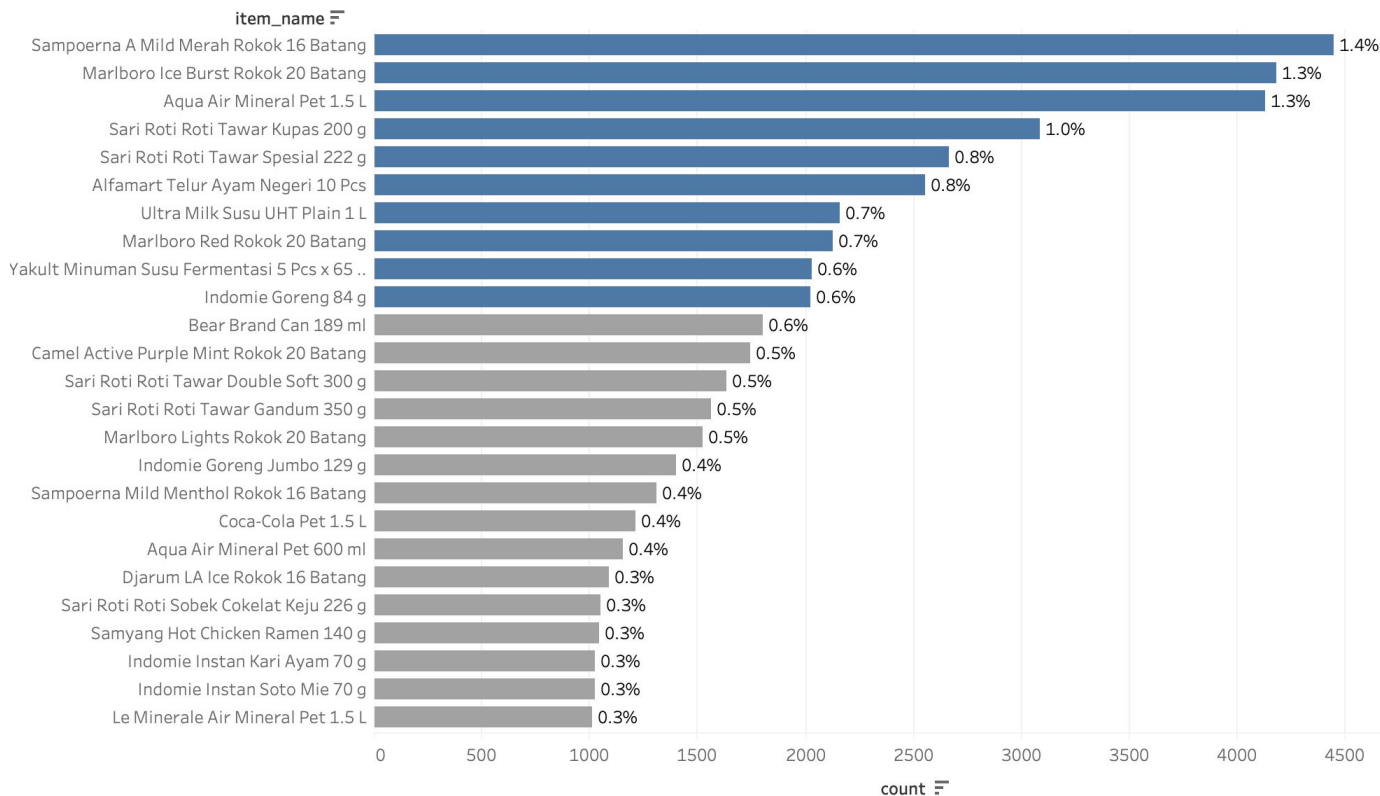
CT Analysis - From iOS only

SKU Item Added

In the last 30 days

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- **ServiceType** contains "Go-Buy" ✗
- AND **Source** contains "Search Screen" ✗

Rokok, Roti are the popular items added to cart after search

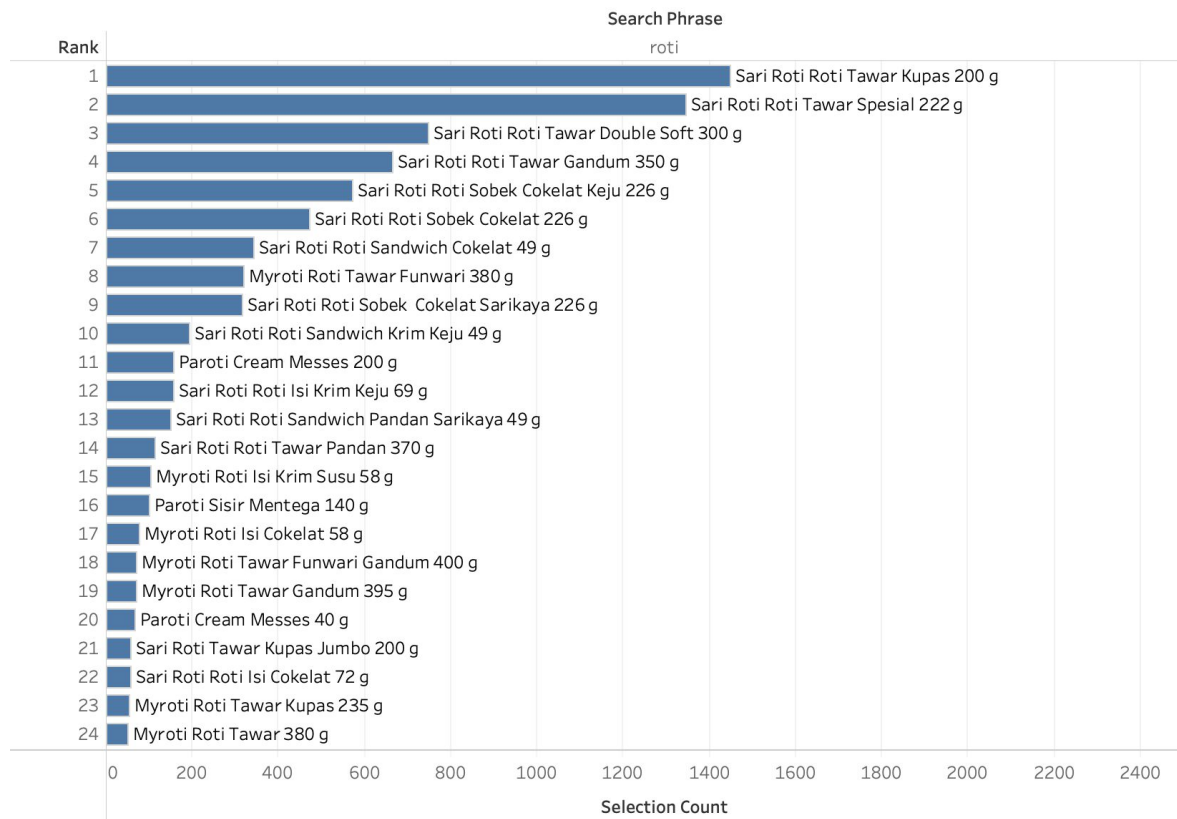


Top 10 items comprise of **9.1%** of the total items

Rest long tail comprising of 4.4k items

[Detailed List](#)

Historical item selections for popular search terms should form an input to improve search relevance



[Detailed List](#)

Recommendations

Recommendations

Short Term:

- Historical list for popular items for top N searches can be used to improve relevance
- This list can be also be broken down by categories if needed

Long Term:

- Create backend search log to track all results shown
- Develop relevance model to enhance personalisation
Eg. rokok for A is Dunhill, while for B is samporna
- Understanding user intent for generic terms to enhance relevance
- Use of [NDCG metric](#) to benchmark search

Important Links

[How GoMart Search Works](#)

93% of searches queries are 3 term searches

- 93% of searches are either with one or term searches
- This implies user is looking to spend minimum effort in searching

